Scaling a Strategy From $20M to $20B: Inertia, Path-Dependent Constraints, and Burn-In

Mathew Thiel

Applied Math and Statistics Master's Student
Stony Brook University, Stony Brook, New York

March 12, 2026

Datasets

Price Data:

- Daily and monthly (adjusted) prices from the CRSP and WRDS[1] [10].

Factor Data:

- Approximately 148 monthly asset-level factors as described in Jensen et. al. [7].
- Factors are constructed from sorts of the top (bottom) 20% of factor scores at a given time.
- Assets are weighted by rolling point-in-time winsorized market cap within each factor.
- Factors are signed according to Jensen et. al. [7].
- In this case, only large and mega-cap (at each point-in-time) stocks were used to construct the factors.

[1] CRSP: Center for Research in Security Prices; WRDS: Wharton Research Data Services

Strategy Flow

1. Select the K-Best (40) factors by Sharpe Ratio (SR) over lookback window i (5 years) with factor return mean μ_{f_i} and variance σ_{f_i}:

$$\mathbf{SR}_{f_i} = \frac{\boldsymbol{\mu}_{f_i}}{\boldsymbol{\sigma}_{f_i}}. \tag{1}$$

2. Shrink factor covariance matrix Σ_f^I using Ledoit-Wolf shrinkage [8].
3. Construct asset covariance matrix from shrunk factor covariance Σ_f^I and factor loadings W_{a_t} at the current period t:

$$\Sigma_a = W_{a_t} \Sigma_f^I W_{a_t} \tag{2}$$

4. Optimize asset weights using maximum SR with path-dependent constraints.
5. Rebalance monthly.

Maximum Sharpe Ratio Optimization

The general maximum Sharpe Ratio optimization problem is as follows [6]:

$$\max_{w \in \mathcal{W}} \quad \frac{\boldsymbol{\mu}_e^\top w}{\sqrt{w^\top \Sigma w}} \tag{3}$$

$$\text{s.t.} \quad \mathbf{1}^\top w = 1, \tag{4}$$

$$w \in \mathbb{R}^N \tag{5}$$

$$(\text{optional } w \geq 0) \tag{6}$$

But this is non-convex!

Maximum Sharpe Ratio Optimization, Convex Formulation

The quadratic reformulation becomes a minimization problem with scaled weights [6]:

$$\arg \min_{(\widetilde{w},\gamma)\in\widetilde{\mathcal{W}}} \widetilde{w}^\top \Sigma\, \widetilde{w} \tag{7}$$

$$\widetilde{\mathcal{W}} = \left\{ (\widetilde{w},\gamma) \in \mathbb{R}^{N+1} \;:\; (\mu - r_f \mathbf{1})^\top \widetilde{w} = 1,\;\; \mathbf{1}^\top \widetilde{w} = \gamma,\;\; \widetilde{w} \geq 0 \right\} \tag{8}$$

$$w^\star \;=\; \frac{\widetilde{w}^\star}{\gamma^\star} \tag{9}$$

Reducing Turnover

L1 norm on weight changes with regularization parameter α [4]:

$$\min_{\widetilde{w},\,\tilde{\gamma},\,\widetilde{w}^{+},\,\widetilde{w}^{-}} \quad \widetilde{w}^{\top}\Sigma\,\widetilde{w} \;+\; \alpha\sum_{i}\left(\widetilde{w}_i^{+} + \widetilde{w}_i^{-}\right) \tag{10a}$$

$$\text{s.t.} \quad (\mu - r_f\mathbf{1})^{\top}\widetilde{w} \;=\; 1 \tag{10b}$$

$$\mathbf{1}^{\top}\widetilde{w} \;=\; \tilde{\gamma} \tag{10c}$$

$$w^{*}_{t-1,i}\,\tilde{\gamma} \;-\; \widetilde{w}_i \;=\; \widetilde{w}_i^{+} - \widetilde{w}_i^{-}, \qquad \forall i \tag{10d}$$

$$\tilde{\gamma} \geq 0, \quad \widetilde{w}_i^{+} \geq 0, \quad \widetilde{w}_i^{-} \geq 0, \qquad \forall i \tag{10e}$$

Path-Dependent Constraints: Capturing Inertia

PDCs capture inertia by keeping signals in the portfolio until a "strong force" (direction change) acts upon the asset [4]:

$$0 \leq \widetilde{w}_i \leq M_i, \qquad \forall i \in I_{t+1}^{+} \setminus A_t^{s} \tag{11a}$$

$$-M_i \leq \widetilde{w}_i \leq 0, \qquad \forall i \in I_{t+1}^{-} \setminus A_t^{\ell} \tag{11b}$$

$$0 \leq \widetilde{w}_i \leq w_{t-1,i}^{*}\,\tilde{\gamma}, \qquad \forall i \in A_t^{\ell} \setminus (I_{t+1}^{+} \cup I_{t+1}^{-}) \tag{11c}$$

$$w_{t-1,i}^{*}\,\tilde{\gamma} \leq \widetilde{w}_i \leq 0, \qquad \forall i \in A_t^{s} \setminus (I_{t+1}^{+} \cup I_{t+1}^{-}) \tag{11d}$$

$$\widetilde{w}_i = 0, \qquad \forall i \in (I_{t+1}^{+} \cap A_t^{s}) \cup (I_{t+1}^{-} \cap A_t^{\ell}) \tag{11e}$$

$$M_i = m_i\,\tilde{\gamma}. \tag{12}$$

Where m_i is the maximum position size for a given asset.

Full Optimization Problem

$$\min_{\widetilde{w},\,\tilde{\gamma},\,\widetilde{w}^+,\,\widetilde{w}^-} \quad \widetilde{w}^\top \Sigma\, \widetilde{w} \;+\; \alpha \sum_i \left(\widetilde{w}_i^+ + \widetilde{w}_i^-\right)$$

$$(\mu - r_f \mathbf{1})^\top \widetilde{w} = 1$$

$$\mathbf{1}^\top \widetilde{w} = \tilde{\gamma}$$

$$w^*_{t-1,i}\tilde{\gamma} - \widetilde{w}_i = \widetilde{w}_i^+ - \widetilde{w}_i^-,\ \forall i$$

$$\tilde{\gamma} \geq 0,\quad \widetilde{w}_i^+ \geq 0,\quad \widetilde{w}_i^- \geq 0,\ \forall i$$

$$w^\star = \frac{\widetilde{w}^\star}{\tilde{\gamma}^\star},\quad M_i = m_i \tilde{\gamma}$$

$$\begin{aligned}
0 \leq \widetilde{w}_i \leq M_i, &\qquad \forall i \in I^+_{t+1} \setminus A^s_t \\
-M_i \leq \widetilde{w}_i \leq 0, &\qquad \forall i \in I^-_{t+1} \setminus A^\ell_t \\
0 \leq \widetilde{w}_i \leq w^*_{t-1,i}\tilde{\gamma}, &\qquad \forall i \in A^\ell_t \setminus (I^+_{t+1} \cup I^-_{t+1}) \\
w^*_{t-1,i}\tilde{\gamma} \leq \widetilde{w}_i \leq 0, &\qquad \forall i \in A^s_t \setminus (I^+_{t+1} \cup I^-_{t+1}) \\
\widetilde{w}_i = 0, &\qquad \forall i \in (I^+_{t+1} \cap A^s_t) \cup (I^-_{t+1} \cap A^\ell_t)
\end{aligned}$$

Burn-In

- Running the optimization at the first timestep would allocate $20 billion to the market all at once.
- To avoid this, I implement a "Burn-in" period before the optimization is allowed to run.
- During the Burn-in period, the strategy buys the maximum % average dollar volume or position size for incoming assets until an investment threshold is reached (70%).
- This spreads market impact out over time and avoids the initial capacity drop that a regular strategy would face.

Large Cap, Burn-in, Reg=0.1 Cumulative Return

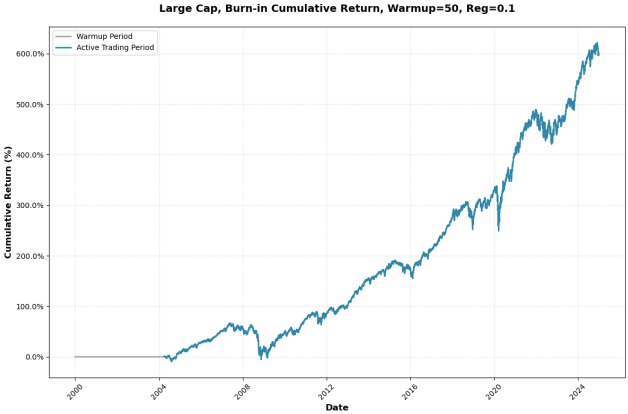

Figure: Large Cap universe with Burn-in cumulative returns with max_pos=0.025.

Large Cap, No Burn-in, Reg=0.1 Capacity

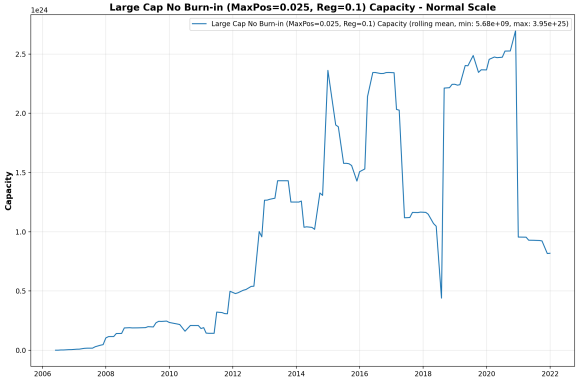

Figure: Maximum amount of wealth able to be invested in the given strategy such that no trade exceeds 5% of the underlying stock's average dollar volume, min=$\$5,680,000,000$, max=$\$3.95 * 10^{25}$.

Large Cap, Burn-in, Reg=0.1 Capacity

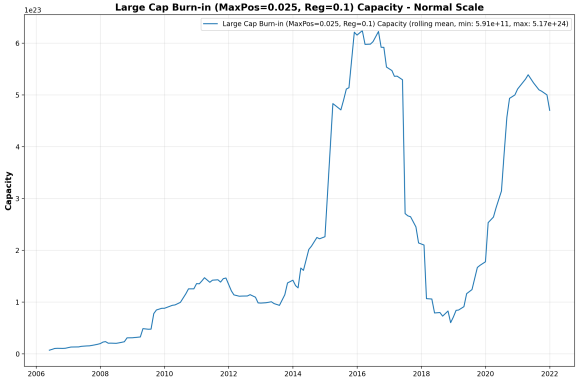

Figure: Maximum amount of wealth able to be invested in the given strategy such that no trade exceeds 5% of the underlying stock's average dollar volume, min=$\$591,000,000,000$, max=$\$5.17 * 10^{24}$.

Turnover of Holding a Single Factor As-is

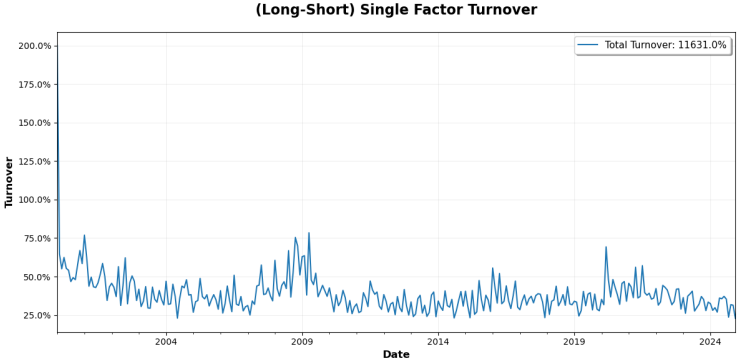

Figure: Turnover of holding a single factor (total=11631%).

Turnover of the Large Cap Universe Strategy Configurations, No Reg. vs. Reg.

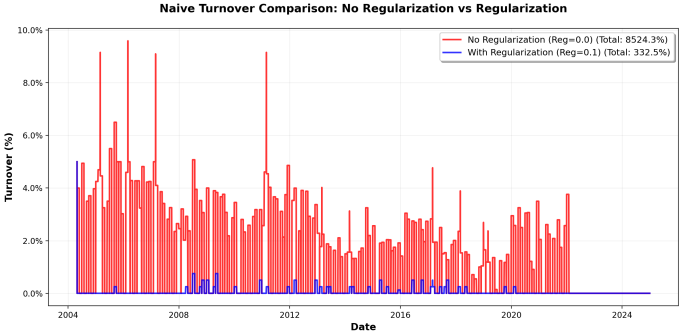

Figure: Turnover Comparison of not regularized (total=8524.3%) vs regularized (total=332.5%) strategies.

Summary Statistics

Metric	Large Cap No Burn-in	Large Cap Burn-in
Total Return (Geometric)	674.98%	597.53%
Annualized Return (Geometric)	8.55%	8.09%
Annualized Volatility (Arithmetic)	12.12%	12.80%
Sharpe Ratio (Annualized)	0.706	0.632
Maximum Drawdown	-38.64%	-43.28%
Minimum Capacity	$5,680,000,000	$591,000,000,000
Turnover	553.8%	332.5%

Table: Performance metrics for Large Cap strategies with and without Burn-in (MaxPos=0.025, Reg=0.1).

Additional Considerations

Transaction Costs without Regularization

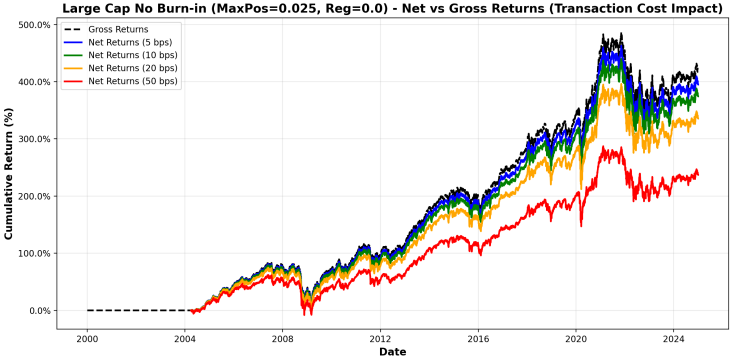

Figure: Cumulative returns of Large Cap no Burn-in no Reg strategy after various fixed transaction costs.

Transaction Costs with Reg=0.1, Burn-in

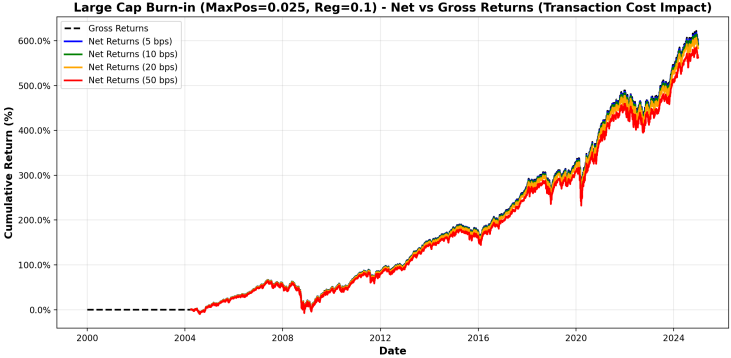

Figure: Returns of Large Cap Burn-in Reg=0.1 strategy after various fixed transaction costs.

Return of the Burn-in Strategy When Executing at Various Prices

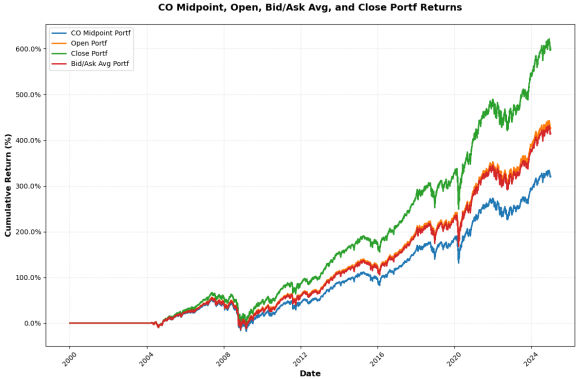

Figure: Cumulative returns of Burn-in strategy when executed at CO (close-open midpoint) price, Bid/Ask average price, open price, and close price.

Appendix: Adding a Factor Risk Model

Factor Risk Model: Factors

- **Currencies**: USD/EUR[2], (Euro), USD/GBP (Great British Pound), JPY/USD (Japanese Yen), CAD/USD (Canadian Dollar), CHF/USD (Swiss Franc), USD/AUD (Australian Dollar).
- **Styles**: Fama French 5 Factor [3]: SMB (Small Minus Big (Size)), HML (High Minus Low (Value)), RMW (Robust Minus Weak (Profitability/Growth)), CMA (Conservative Minus Aggressive (Investment)).
- **Market**: Market factor provided by Fama-Fench [3].
- **Industries**: 17 Industry portfolios classified by SIC code, provided by Fama-French.
- **Total: 28 factors**

[2] The Euro was not officially introduced as an accounting currency until 1999. To create the full history, the Deutsche Mark, normalized by the fixed conversion rate of 1.95583, was spliced with USD/EUR up to the introduction date.

Factor Risk Model: Data

- Currencies are provided by St. Louis Fed Federal Reserve Economic Data [1].
- Style attributes used to construct factor returns are provided by WRDS, created using the pyanomaly package [5].
- Preprocessing:
  - Monthly returns below -50% are set to NaN; above 200% are clipped to 200%.
  - Monthly returns outside of $3 \cdot 1.4826$ (analogue for 3 standard deviations) $\cdot$ median absolute deviation (point-in-time) are set to NaN.
  - $\text{MAD} = 3 \cdot 1.4826 \cdot \text{median}\left(|x_i - \text{median}(X)|\right)$
  - All factors are in log-return space and excess of the risk-free rate (aside from currencies) provided by Fama-French.

Factor Risk Model: Regression Methodologies

- Time series regression: estimate factor betas given factor returns.
  - Rolling Regularized Huber regression [2] with backward elimination
  - Rolling window: 60 months (55 months minimum per window)
  - Exponential weighting: half-life of 36 months in variance space (approx. 72 months in standard deviation space)
  - Regularization: $\alpha = 0.007$ ($\alpha = 0.07$ for currencies); default $\epsilon = 1.35$
  - **For Industries** Map SIC code to industry portfolio; regress on that industry by itself with significance, then backward eliminate the rest of the industry portfolios. Ensures the variance captured by the main industry is not spread among other industries.
- Cross-sectional regression: estimate factor returns given betas/attributes
  - Samples are weighted by the square root of market capitalization (see [9])
  - Attributes (betas) are start of month to keep the regression point-in-time

Factor Risk Model: Dealing with Multicollinearity

- **"Exploding Betas"** Problem - Model cannot distinguish between factor effects
- Classic sign: large betas that flip signs between correlated factors
- Backward Elimination:
  1. Start with all factors
  2. Run the regression
  3. Drop any factors with absolute t-statistic < 2
  4. Repeat until all remaining have $|t| > 2$ or no factors remaining
- VIF (Variance Inflation Factor) $1/(1 - R^2)$:
  - Measures how much a given beta is increased due to multicollinearity
  - Used in currency factor regression before the t-statistic test each round
  - Same process as t-statistic, but drop maximum VIF > 10 for the round

Factor Risk Model: Regression Order

Currencies $\longrightarrow$ Styles $\longrightarrow$ Market $\longrightarrow$ Industries

1. Currencies:
   - Currencies first essentially isolates "local return."
   - If a stock like Honda were in our universe, it would have high exposure to the Yen
2. Styles:
   - Since market and styles are highly correlated, whichever is regressed first will "soak up" market risk
   - If we regressed the market first, the leftover style risk would be tiny. Generally, managers prefer to see style effects.
3. Market:
   - Market regression should now remove "pure" market risk
4. Industries:
   - For industry-specific risk not captured by the market/style regressions.

Factor Risk Model: Updated Optimization

Now that we have a risk model, we modify the original optimization:

- **Orthogonalize alpha factors**: Alpha factors are likely correlated with risk factors; orthogonalizing ensures that risk drivers are not used to select assets.
  - *Returns*: Regress (OLS) each alpha factor return on risk factor returns; use residuals as "purified" alpha returns.
  - *Loadings*: Regress each alpha factor loading (asset–factor mapping) on risk betas; use residuals as "pure" alpha betas $\widetilde{W}_t$.
- **Mean** (expected returns): Now comes from the orthogonalized alpha weights/factor sample mean.
- **Covariance**: Use the risk model factors and idiosyncratic risk (Δ residual variance):

$$\Sigma \;=\; B\,\Sigma_f\,B^\top \;+\; \mathrm{diag}(\Delta),$$

with B = risk betas (assets $\times$ risk factors), Σ_f = risk factor covariance (shrunk using Ledoit-Wolf [8] shrinkage), and point-in-time residual variance per asset.

Cumulative Returns: No Risk Model

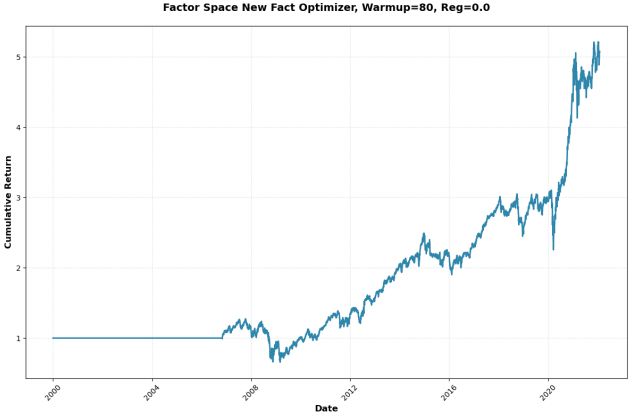

Figure: Cumulative returns (no risk model).

Cumulative Returns: Risk Model

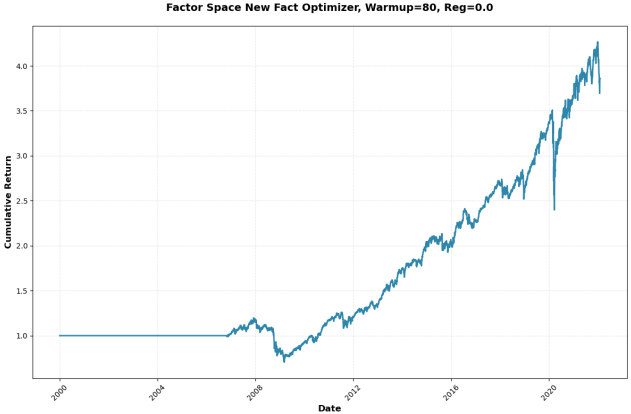

Figure: Cumulative returns (risk model).

Asset Counts: No Risk Model

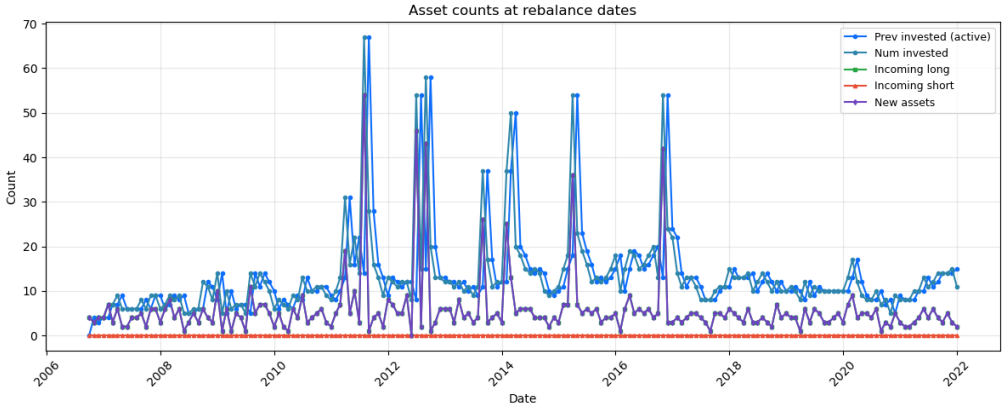

Figure: Asset counts (no risk model).

Asset Counts: Risk Model

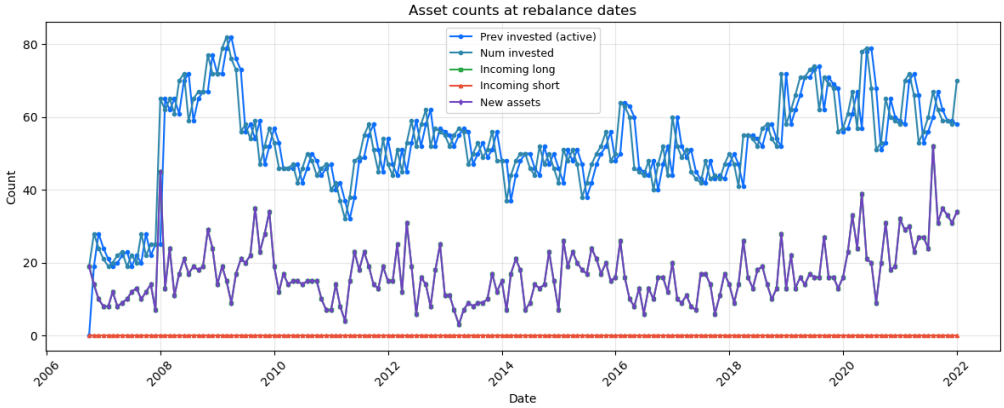

Figure: Asset counts (risk model).

Turnover: No Risk Model

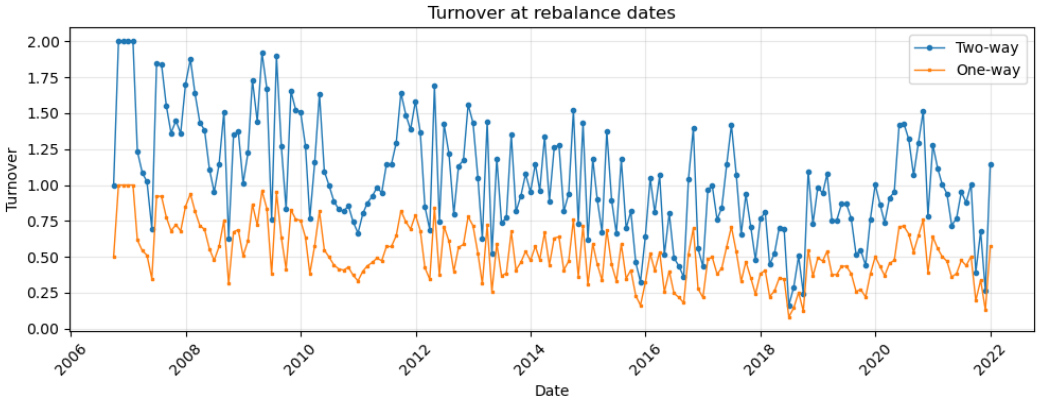

Figure: Turnover (no risk model).

Turnover: Risk Model

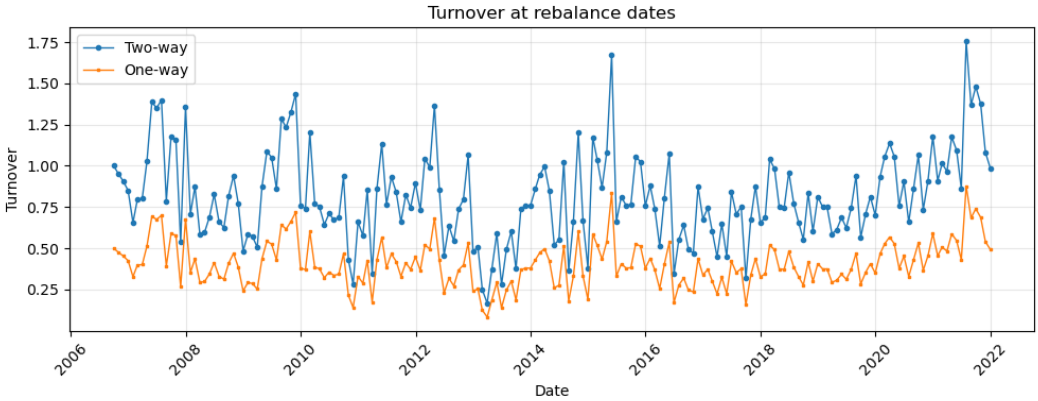

Figure: Turnover (risk model).

Turnover Breakdown: No Risk Model

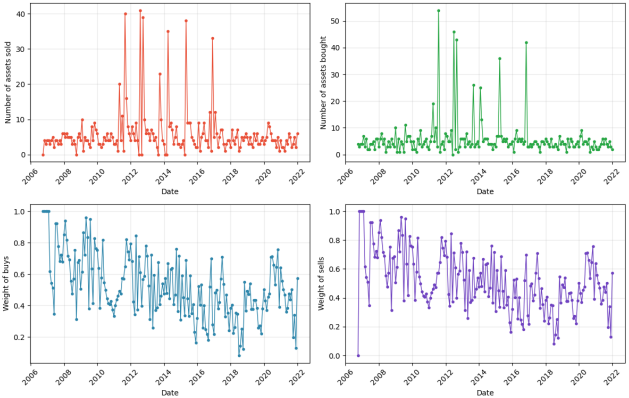

Figure: Turnover breakdown (no risk model).

Turnover Breakdown: Risk Model

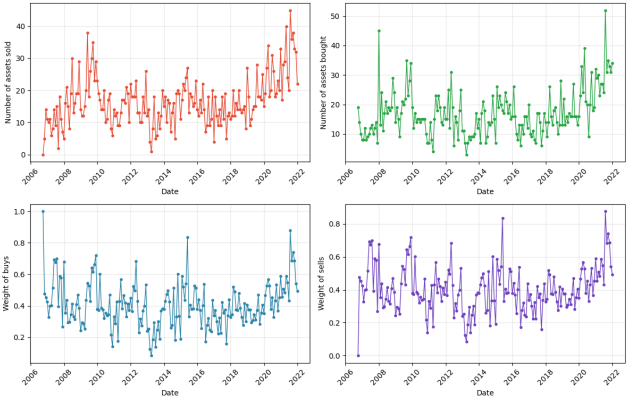

Figure: Turnover breakdown (risk model).

Summary Statistics

Table: Summary statistics (monthly rebalancing)

Metric	Risk model	No risk model
Annualized volatility	**0.1289**	0.1805
Total cumulative return	2.8597	4.0770
Sharpe ratio	**0.5400**	0.4988
Max drawdown	0.4121	0.4847
Total turnover (one-way)	74.7747	95.3954
Average turnover (one-way)	0.4064	0.5185
Num rebalances	184.0	184.0

Risk Model: 28.5% reduction in Annualized Vol, 21.6% reduction in total turnover

References

[1] Federal Reserve Economic Data — FRED — St. Louis Fed.

[2] HuberRegressor.
https://scikit-learn/stable/modules/generated/sklearn.linear_model.HuberRegressor.html.

[3] Kenneth R. French - Data Library.

[4] S. Chitsiripanich, M. S. Paolella, P. Polak, and P. S. Walker.
Smoothing Out Momentum and Reversal, Aug. 2024.

[5] chulwoohan.
Chulwoohan/pyanomaly, Feb. 2026.

[6] W. Deng, P. Polak, A. Safikhani, and R. Shah.
A Unified Framework for Fast Large-Scale Portfolio Optimization.

[7] T. I. Jensen, B. T. Kelly, and L. H. Pedersen.
Is There a Replication Crisis in Finance?

[8] O. Ledoit and M. Wolf.

Contact Information

Feel free to reach out to me with any questions:

- Email: mathew.thiel@stonybrook.edu
- LinkedIn: mathew-thiel